

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
										S	T	U	V	X	Y	Z	
										18	19	20	21	22	23	24	25

Assignment - 261100690022

1. Encrypt "CHASE YOUR DREAMS"

a) Ceaser Cipher:

CHASE YOUR DREAMS

$$\Rightarrow E = (P + 3) \bmod 26$$

$$C \rightarrow 2 + 3 = 5 \bmod 26 \Rightarrow F$$

$$H \rightarrow 7 + 3 \bmod 26 \Rightarrow K$$

$$A \rightarrow 0 + 3 \bmod 26 \Rightarrow D$$

$$S \rightarrow 18 + 3 = 21 \Rightarrow V$$

$$E \rightarrow 4 + 3 \bmod 26 \Rightarrow H$$

$$Y \rightarrow 24 + 3 \bmod 26 \Rightarrow B$$

$$O \rightarrow 14 + 3 \bmod 26 \Rightarrow R$$

$$U \rightarrow 20 + 3 \bmod 26 \Rightarrow X$$

$$R \rightarrow 17 + 3 \bmod 26 \Rightarrow V$$

$$D \rightarrow 3 + 3 \bmod 26 \Rightarrow G$$

$$M \rightarrow 12 + 3 \bmod 26 \Rightarrow P$$

$$S \rightarrow 18 + 3 \bmod 26 \Rightarrow V$$

$$E \rightarrow H$$

$$A \rightarrow D$$

$$M \rightarrow P$$

$$S \rightarrow V$$

$E \Rightarrow FKDYH BRXUGUHDPV$

b) Shift Cipher : Key = 7

$$E = (P + 7) \bmod 26$$

$$C = (2 + 7) \bmod 26 = 9 \bmod 26 = 9 \Rightarrow J$$

Similarly, CHASE YOUR DREAMS
Cipher $\rightarrow$ J O H Z L F V B Y K Y I I H T Z

c) ROT13

$$E = (P + 13) \bmod 26$$

CHASE YOUR DREAMS

2 7 0 18 4 24 14 20 17 3 17 4 0 12 18 $\rightarrow P$

15 20 13 31 17 37 27 33 30 16 30 17 13 25 31 $+13$

$(C \Rightarrow) P W N F R L B H E Q E R N Z F$ $\bmod 26$

Decryption:

1) WRWDOOB VHFUHW PHVVDJH

ceaser cipher

$$D = (E - K) \bmod 26 \quad K = 3$$

shift 3 letters back

$$W = (22 - 3) \bmod 26 = 19 = T$$

TOTALLY SECRET MESSAGE

b) Shift Cipher: 5

$$D = (E - 5) \bmod 26$$

$$m = 12 - 5 = 7 \bmod 26 \Rightarrow H$$

PT $\Rightarrow$ M J Q Q T B T W Q I
 $\downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow$
 H E E L O W O R L D

c) S J B D T W P Y N R I

$$K = 5$$

$\downarrow \downarrow \downarrow$

$$P = (E - 5) \bmod 26$$

NEW YORK TIME

3) ROT 13

F R A Q Z B E R P B S S R R

$$PT = (E - 13) \bmod 26$$

$$F = 5 - 13 \bmod 26 \Rightarrow 5$$

$$P = 17 - 13 \bmod 26 \Rightarrow E$$

$$E = 4 - 13 \bmod 26 \Rightarrow R$$

$$A = 10 - 13 \bmod 26 \Rightarrow N$$

$$P = 15 - 13 \bmod 26 \Rightarrow Q$$

$$Q = 16 - 13 \bmod 26 \Rightarrow D$$

$$S = 18 - 13 \bmod 26 \Rightarrow F$$

$$Z = 25 - 13 \bmod 26 \Rightarrow M$$

$$B = 1 - 13 \bmod 26 \Rightarrow Q$$

$\Rightarrow$ SEND MORE COFFEE